Ryan Vincent

Boston, MA
VincentRyanBaker@gmail.com | github.com/CoraBeanz

Overview: I enjoy working on novel applications of software, hardware, optics, & materials as well assisting others in their quests to do so. In my free time I develop ROS based optical benches and hyperspectral robotic perception.

PROFESSIONAL EXPERIENCE

Optical Systems Engineer Feb 2026 - Present
KineoLabs (Start-Up), Billerica, MA

Overview: *Bringing to market affordable, compact, wavelength-sweeping laser engines for the spectroscopy, telecom, & medical imaging markets.*

Notable Personal Contributions:

- Designed, developed, & maintained full stack, cross platform applications (FE/BE/APIs/SDKs) for multiple clients on Windows OS and Linux based single board computers (Nvidia Jetson Nano) while managing deployment security.
- Developed company's core software IP: computer vision for Fourier domain OCT, speckle variance angiography, machine learning based diagnostics for ophthalmic pathology, and GPU (CUDA) based data processing algorithms.
- Lead mechanical, electrical/PCB, & embedded systems design for the company's upcoming ophthalmic imaging system with Optical Coherence Tomography (OCT) and Fundus imaging modalities.
- Created a full digit twin for our ophthalmic imaging system combined with multiphysics simulation to develop and perfect new imaging modalities through reinforcement-learning and Bayesian optimization.
- Developed & simplified optomechanical systems & kinematic fixturing for low-cost manufacturing & high volume assembly without sacrificing tolerances.

Relevant Patents: Free-space multimodal ophthalmic imaging system combining line-field swept source OCT and line scanning laser ophthalmoscopy **Provisional Application No. 64/077,487**

Freelance September 2025 – March 2026
Independent, Boston

- **Luminera** - Assisting a research group trying to spin out a startup from BU/OHSU. Designing electronic, photonic, and plasmonic biosensors and building out physics informed neural networks to detect single molecule time-transient signals from chromophores bound to peptides with the goal of protein sequencing.
- **ChromatiQ** - Working with a group out of Harvard to spin out a startup based on a machine learning/diffractive optic based hyperspectral imaging sensor technology for low cost hyper/multispectral robotic imaging sensors.

Senior Materials Scientist (4th Employee) June 2023 – May 2025 (Left to start PhD)
Irradiant Technologies (Start-Up), Cambridge, MA

Overview: *Startup out of the MIT Media Labs & Activate Fellowship. Developing a new method of ultrafast, nanoscale 3d lithography technique using spatiotemporally structured pulsed laser light for creating nanophotonic & nanoelectronic devices inside nanoporous scaffolds.*

Notable Personal Contributions:

- Led development of the company's nanoporous ceramic wafers & multiphoton-reactive polymeric and inorganic nanomaterials for lithography.
- Advised on, wrote proposals, & managed material science technical deliverables for customer projects, SBIRs, & government grants for the Army, Navy, DARPA, DOD, and NIST.
- Developed software/hardware for lithography systems, opto-mechanical control systems, machine learning based automated nanophotonic metrology systems, & automated nanoparticle spectroscopy.
- Built out initial chemical wet laboratories, designed multistage chemical reactors, liquid-handling robotics (Opentron, custom pneumatic systems), pilot-scale production, & automation.

Relevant Patents: *Systems and methods for integrated free space optical elements and devices WO2025235812A1*

Process Engineer (3rd Employee), December 2021 – March 2023 (Dissolved)
RadiantNano (Start-Up), Framingham, MA

Overview: *Startup out of the Activate Fellowship. Developing a new class of completely solution processed perovskite scintillators with wavelength tunable emission centers for applications in imaging, x-ray/γ-rays spectroscopy, and nuclear batteries.*

Notable Personal Contributions:

- Led process development and scale-up for various quantum dot nanoparticle, perovskite syntheses, and liquid phase heteroepitaxial incorporation of the quantum dot color centers into the bulk perovskite crystals.
- Built out scintillator and nanomaterial metrology pipelines (light yield, spatial, spectral resolution, simulations) .
- Developed wavelength-tunable scintillators of various form-factors (flexible polymer nanocomposites, sintered polycrystalline cubes for spectroscopy, & inkjet pixelated arrays, single crystals).
- Design and build-out complete laboratory fully equipped with a chemistry wet-lab space, material processing space, and a visible light/x-ray/gamma ray spectroscopy characterization space.

Materials Engineer (9th Employee), February 2020 – September 2021
DUST Identity (Start-Up), Framingham, MA

Overview: *Startup out of the MIT Media Labs. Using embedded fluorescent diamond nanoparticle ensembles as unclonable security tags to link high value physical assets to their digital records.*

Notable Personal Contributions:

- Developed techniques for improving fluorescence & optical properties of the company's core diamond nanoparticle technology.
- Built a portfolio of diamond nanoparticle-integrated products: inks, aerosolized conformal coatings, thin films, microchip epoxy molding compounds, water-soluble/tamper-resistant films.
- Integrated the technology into industrial-scale manufacturing lines (reel-to-reel, sticker printing, PCB coating, inkjet, electrostatic printing), taking products from prototype to on-site customer demonstrations.

Relevant Patents: *Producing Identity Markers by Electrostatic Printing 20250060689*

EDUCATION

Master of Science in Physics, January 2018 - December 2019
Marshall University, Huntington, WV

Research (Graduate Research Assistant), January 2018 – December 2019

- Studied the nanoparticle synthesis kinetics and self-assembly of 2D nanoparticle monolayers at the liquid-vapor interface. Fully funded through the NSF, the self-assembled nanoparticle structures were then investigated for application as electrostatically charged, ultra-thin, high flux water filtration membranes.

Master's Thesis: *https://mds.marshall.edu/etd/1268/*

Other Experience: *Physics 1 lab instructor. Physics and department outreach. Assisted with hosting fractography and biological specimen electron microscopy workshops.*

Bachelor of Science in Physics, September 2013 - May 2017
Bloomsburg University, Bloomsburg, PA

Research (Undergraduate), 2016
Bloomsburg University Astrophysics Lab, Bloomsburg, PA

- Adopting non-linear analysis techniques, analyzed the stochastic brightness variations of hundreds of stars, to determine various qualities such as asteroseismological fluctuations and exoplanet discoveries.

Research (Undergraduate), 2015
Bloomsburg University Ultra-Cold Lab, Bloomsburg, PA

- Worked on developing custom laser control systems used in ultra-cold atom cooling and trapping to achieve a Bose-Einstein Condensate.